

Does Weight Sharing Reduce Stereotype Association? An Iso-Loss Comparison of Looped Transformer Encoders

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Abstract

A pre-trained encoder is built once and then reused, so any stereotypical association it holds is passed on to every system that adopts it. Encoders that apply one block of weights at several depths, known as looped or recursive models, are attractive because they keep effective depth at a lower storage cost. Whether such reuse also changes what a model records about social groups is not known, and the question is awkward to answer: a shared model and an unshared model of the same size do not reach the same language modelling quality, and a better trained model gives sharper answers on every probe. This study compares four encoders at matched validation loss rather than at matched parameters or matched tokens, so that a difference in measured association cannot be explained by one model having learned more language. On a paired stereotype benchmark of 1508 items, the two weight-shared encoders record a lower stereotype effect size than the unshared baseline, and both contrasts survive correction for multiple comparisons. Adding parallel streams and hyper-connections on top of looping gives no further reduction and makes training less stable. The same models answer at chance on a coreference instrument, so the result is reported as a statement about association in the masked language modelling head at this training scale, and not as evidence about downstream behaviour.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; *Learning paradigms*; • **Social and professional topics** → *User characteristics*.

Keywords

social bias, masked language models, parameter sharing, looped transformers, fairness evaluation, iso-loss comparison

1 Introduction

Pre-trained encoders sit underneath much of current retrieval and text classification practice. One masked language model is trained and then reused across many tasks, so a stereotypical association acquired during pre-training is inherited by every system built on top of it. Which design choices change these associations is therefore a question about shared infrastructure rather than about a single application.

Weight sharing across depth is one such choice. ALBERT ties a single transformer block across all layers and reaches competitive quality at a fraction of the storage [10]. Recent work applies a small block repeatedly to obtain effective depth without a matching parameter count, and reports gains on reasoning tasks [1, 16]. The motivation offered for these designs is efficiency. A separate line of work finds that changing model capacity changes what a model records about social groups: pruning and quantisation shift error rates unevenly across demographic groups [7, 15, 21]. Read together,

these two lines raise a question that neither answers. If a looped encoder must cover twelve depths of computation with six blocks of weights, does the pressure to reuse those blocks suppress the narrow item-level associations that stereotype benchmarks detect?

A confound stands in the way of an answer. A shared encoder and an unshared encoder with the same parameter count do not reach the same language modelling quality. A model further along in training also gives sharper predictions on every probe, stereotype probes included. Comparing two architectures after a fixed number of training tokens therefore measures training progress together with architecture. Existing comparisons of compression and fairness do not control for this directly, which makes their results hard to attribute to one cause.

This work removes the confound by matching on outcome rather than on budget. Each encoder is trained until its validation loss crosses a list of values fixed in advance, and a snapshot is written at each crossing. Encoders are then compared only at snapshots of equal validation loss. Under this protocol a difference in measured association cannot be explained by one model having learned more of the language.

Four encoders are compared: an unshared baseline; a looped encoder that applies six blocks twice; an ALBERT-style encoder that applies one block twelve times; and a looped encoder extended with four parallel residual streams joined by hyper-connections [24]. The last of these tests whether routing capacity added on top of sharing changes the picture.

This paper makes three contributions.

- (1) An iso-loss protocol that attributes a difference in measured stereotype association to architecture rather than to training progress, together with a capability gate that states when the measurement is interpretable.
- (2) A controlled comparison of four encoders under this protocol, in which both weight-shared encoders record a lower stereotype effect size than the unshared baseline at matched validation loss.
- (3) A negative result: parallel streams and hyper-connections added on top of looping give no further reduction, and make optimisation less stable.

Section 2 reviews prior work on bias measurement and on weight sharing. Section 3 defines the encoders, the matching protocol and the measures. Section 4 gives the training setup and Section 5 reports results. Section 7 states what the evidence does not support.

2 Related Work

Measuring stereotype association in masked models. CrowS-Pairs scores a model on minimal pairs that differ only in a demographic term and records how often the stereotypical member receives the higher pseudo-log-likelihood [12]. StereoSet uses a related paired construction with an added language modelling score [11]. Kurita et al. [9] introduced the masked-position probing that both

build on. The instruments have known weaknesses. Blodgett et al. [2] audited these benchmarks and found unclear or inconsistent items in a large share of pairs, and warned against reading one aggregate number as a measure of harm. Wang et al. [18] separate correlation-type benchmarks, which record association, from descriptive and normative benchmarks, which record whether a model treats groups differently when it should. The measure used here is of the correlation type, and Section 7 states what follows from that.

Weight sharing and looped depth. ALBERT ties one encoder block across all layers and trades quality for storage [10]. Saunshi et al. [16] show that a k -layer block looped L times approaches the reasoning performance of a kL -layer model, and argue that depth rather than parameter count drives several reasoning tasks. Bae et al. [1] learn a per-token recursion depth over a shared block. Geiping et al. [6] scale a recurrent-depth model to test-time compute, and Zhu et al. [25] and Frey et al. [5] study related looped language models. Hyper-connections generalise the residual connection to several parallel streams with learned mixing [24], later constrained to a manifold by Xie et al. [20], and Zeitoun et al. [22] combine looping with hyper-connections. This body of work evaluates quality, efficiency and reasoning. It does not report the effect of sharing on social bias.

Capacity, compression and fairness. Hooker et al. [7] found that pruning degrades accuracy unevenly across groups, with a small subset of examples absorbing most of the damage. Xu and Hu [21] and Ramesh et al. [15] report mixed outcomes for compression and fairness in language models, with the direction depending on the compression method and the benchmark. Voria et al. [17] localise stereotype-carrying neurons inside pre-trained transformers. These studies compress a model after training. The present work varies sharing before training and compares at matched quality, which separates the architectural question from the recovery dynamics of a compressed checkpoint.

3 Method

3.1 Architectures

All four encoders use hidden size 768, twelve effective depths, one tokeniser and the same masked language modelling objective [4]. They differ only in how weights are reused across depth.

VanillaBERT holds twelve independent blocks and is the unshared control. *LoopedBERT* holds six blocks and applies the stack twice, halving the depth-unique parameters. *ALBERTLoopedBERT* holds one block and applies it twelve times, the extreme point of the sharing axis [10]. *HyperloopBERT* keeps the six-block loop and adds four parallel residual streams joined by hyper-connections [24]. Instead of a single residual path, the model carries n streams and mixes them at each depth with a learned matrix,

$$\mathbf{H}^{(\ell+1)} = \mathbf{A}^{(\ell)} \mathbf{H}^{(\ell)} + \mathbf{B}^{(\ell)} f(\mathbf{H}^{(\ell)}), \quad (1)$$

where $\mathbf{H}^{(\ell)} \in \mathbb{R}^{n \times d}$ stacks the n stream states of width d at depth ℓ , f is the shared transformer block, $\mathbf{A}^{(\ell)} \in \mathbb{R}^{n \times n}$ is a learned matrix that mixes the streams, and $\mathbf{B}^{(\ell)} \in \mathbb{R}^{n \times n}$ distributes the block output back across them. Setting $n = 1$ with $\mathbf{A} = \mathbf{B} = \mathbf{I}$ recovers the ordinary residual connection. Table 1 gives the resulting sizes.

Table 1: The four encoders. All use hidden size 768 and twelve effective depths, and differ only in weight reuse across depth.

Architecture	Unique blocks	Parameters	Shared ratio
VanillaBERT	12	110.1 M	0.00
LoopedBERT	6	67.6 M	0.50
HyperloopBERT	6	86.5 M	0.50
ALBERTLoopedBERT	1	32.1 M	0.92

3.2 Iso-loss matching

The comparison is defined on outcome rather than on budget. A list of target validation loss values is fixed before training. Validation loss is measured at a regular interval during a run, and the first time it falls below a target the model state is written to disk as the snapshot for that band. Encoders are then compared only at snapshots that carry the same band.

Two properties of the protocol affect how the results are read. Several bands are sometimes crossed inside one validation interval, in which case they share a single snapshot; the analysis therefore treats a distinct validation loss, not a band label, as the unit of measurement. Matching is also approximate, since a snapshot is taken at the first measurement below the target and the realised loss sits a little below the nominal value. The realised spread at the comparison point is reported in Section 5.

3.3 Stereotype measure

Each benchmark item is a pair of sentences differing only in a demographic term, one member marked stereotypical and the other anti-stereotypical [12]. For a sentence s with tokens $t_1 \dots t_T$, the pseudo-log-likelihood masks one position at a time and sums the resulting log probabilities,

$$\text{PLL}(s) = \sum_{i=1}^T \log P(t_i | s_{\setminus i}), \quad (2)$$

where $s_{\setminus i}$ is the sentence with position i replaced by the mask token and P is the output of the masked language modelling head. Scoring runs in 32-bit precision, because reduced precision reorders closely matched pairs.

For a pair, the effect size is the length-normalised difference $(\text{PLL}(s_{\text{stereo}}) - \text{PLL}(s_{\text{anti}})) / T$, and the reported statistic is its mean over items with a bootstrap confidence interval. Zero means the model has no preference between the two members. Encoders are contrasted item by item, so each pair contributes one paired difference and the test is a paired permutation test, corrected across the three contrasts by the Holm procedure. Permutation p -values follow Phipson and Smyth [14], which avoids reporting an exact zero.

3.4 Capability gate

A model that predicts almost nothing returns near-tied pseudo-log-likelihoods and so a small effect size, which would be indistinguishable from a genuinely unbiased model. The protocol therefore fixes in advance three conditions under which a bias reading is treated as interpretable. The model must score above chance on a set of GLUE tasks [19]. The effect-size interval of the unshared baseline

must exclude zero. The model must also score above chance on the pro-stereotype split of a coreference instrument [23], which tests whether it can resolve the pronoun at all. Section 5.3 reports the outcome of this gate, which governs how the main result should be read.

4 Experimental Setup

All four encoders are pre-trained on the same 7 billion tokens of FineWeb-Edu [13] at sequence length 128, with a tokeniser fitted once and reused, so no encoder sees a different corpus or segmentation. The corpus is pre-tokenised into fixed-size blocks and stored as a memory-mapped array, and an integrity check compares the tokeniser checksum and block count against a manifest at the start of every run. Training uses AdamW at a peak learning rate of 3×10^{-4} , batch size 512, warmup over a tenth of the steps, gradient clipping at 1.0, and bfloat16 autocast with FlashAttention [3]. Runs execute on one H100 GPU.

HyperloopBERT is the single exception to the shared learning rate. At 3×10^{-4} it diverged to a non-finite loss, and at half that value it diverged later in training; Section 5.6 reports both events. Its released snapshots come from the run at 1.5×10^{-4} . Holding the learning rate fixed across all four would have compared three stable models against one broken configuration. The peak learning rate is therefore tuned per architecture, and the arms are matched on validation loss rather than on hyper-parameters.

Results come from one seed. This constrains the strength of the claims and is taken up in Section 7.

5 Results

5.1 Language modelling quality

Quality falls as weight reuse rises, and the cost is far from linear. Halving the depth-unique blocks from twelve to six costs little, whereas collapsing six blocks to one costs several times more. Figure 1 shows the trajectories. VanillaBERT and LoopedBERT lie on top of each other for most of training and separate only slightly at the end, while ALBERTLoopedBERT stays clearly above both. LoopedBERT is therefore not under real capacity pressure at this scale; ALBERTLoopedBERT is.

Figure 1 also shows a property the tables cannot. HyperloopBERT leaves the early plateau, where a masked language model predicts little beyond token frequency, after roughly half a billion tokens, while VanillaBERT and LoopedBERT stay on it until about 1.4 billion. It reaches the comparison point with about a quarter fewer tokens than the unshared baseline. The same run later diverges.

5.2 Stereotype association at matched quality

Figure 2 plots the stereotype effect size against realised validation loss. Two patterns stand out. For every encoder the effect size grows as training proceeds, so a better model records a stronger stereotype preference: the association is acquired along with the language. At the lowest-loss end the unshared baseline separates upward from the two shared encoders.

Table 2 reports the comparison point. The four snapshots are matched to within 0.034 nats, so the remaining differences cannot be attributed to one model being better trained. LoopedBERT and

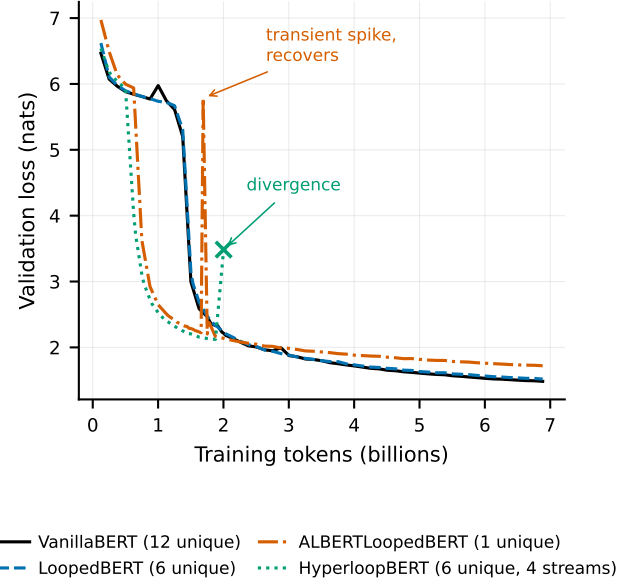


Figure 1: Validation loss against training tokens. HyperloopBERT leaves the plateau earliest and later diverges. ALBERTLoopedBERT shows one transient spike that recovers.

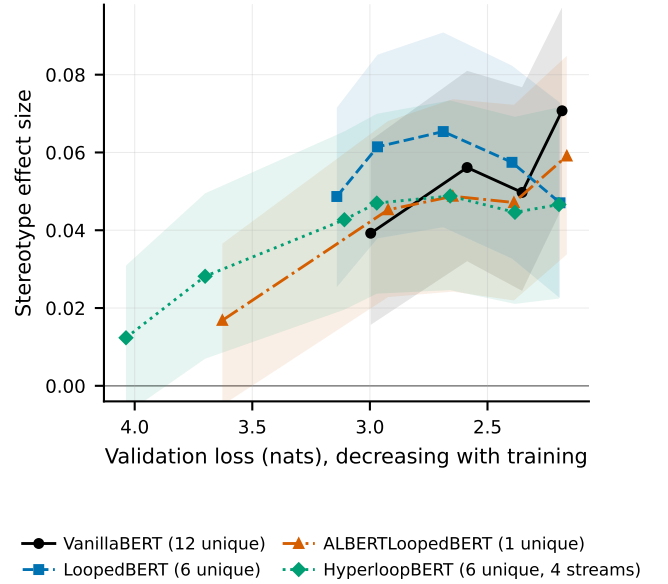


Figure 2: Stereotype effect size against realised validation loss with bootstrap confidence intervals. Effect size grows as loss falls for all four encoders.

HyperloopBERT record the lowest effect sizes, VanillaBERT the highest, and ALBERTLoopedBERT sits between them.

The contrasts fixed in advance appear in Table 3. Both comparisons against the unshared baseline hold after Holm correction, and the comparison between the two shared encoders does not. Weight sharing goes with a lower stereotype effect size at matched quality,

Table 2: Comparison point. All four snapshots lie within 0.034 nats of one another. Effect size carries a bootstrap interval over 1508 paired items.

Architecture	Val. loss	Effect size	95% interval
VanillaBERT	2.183	0.071	[0.046, 0.097]
LoopedBERT	2.192	0.047	[0.023, 0.073]
ALBERTLoopedBERT	2.163	0.059	[0.034, 0.085]
HyperloopBERT	2.196	0.047	[0.022, 0.072]

Table 3: Contrasts fixed in advance, paired permutation test over 1508 items with Holm correction.

Contrast	Δ	p_{Holm}	Cohen’s d	Holds
Vanilla vs. Looped	0.0237	0.0003	0.115	yes
Vanilla vs. Hyperloop	0.0241	0.0004	0.098	yes
Looped vs. Hyperloop	0.0005	0.4668	0.002	no

and the extra machinery in HyperloopBERT adds nothing beyond what plain looping already provides. The paired effects are small, close to a tenth of a standard deviation.

5.3 The capability gate is not met

The gate of Section 3.4 is not satisfied. On the coreference instrument all four encoders answer at chance, with pro-stereotype accuracy between 0.489 and 0.503 and a pro-minus-anti gap within 0.03 of zero throughout. A model that cannot resolve the pronoun cannot show a stereotypical preference in resolving it, so this leg returns no evidence in either direction. The GLUE leg did not yield usable numbers within the available compute and is likewise unmet.

The second leg is met: the effect-size interval of the unshared baseline excludes zero at the comparison point, as Table 2 shows. The consequence is that the architecture contrast is a statement about pseudo-log-likelihood association in the masked language modelling head, not a claim about behaviour on a downstream task. Stating this openly is preferable to presenting the contrast as a general fairness result.

5.4 What the models predict

Aggregate scores hide what a model actually returns, so Table 4 reports masked-position output at the comparison point as log-odds in favour of the masculine pronoun. Two readings follow. The occupational association is visible directly, and it is strongest on the scientist probe for every encoder. The secretary probe then shows why the paired design is needed: the common stereotype runs towards the feminine pronoun, yet every encoder prefers the masculine one, because raw pronoun frequency dominates at this scale. A single probe would mislead, and only the paired difference isolates the demographic term.

5.5 Why the extra streams do not help

The null result invites a mechanistic question: do the four streams carry different information at all? If they collapse onto one another, the extension is inert by construction and the null is uninformative. Two measurements address this.

Table 4: Masked-position log-odds in favour of “he” at the comparison point. A positive value prefers the masculine pronoun.

Probe	Vanilla	Looped	ALBERT	Hyperloop
The doctor ...	+1.47	+2.33	+1.30	+0.46
The secretary ...	+3.00	+2.60	+3.80	+2.71
The scientist ...	+4.22	+2.76	+3.65	+2.93
The teacher ...	−1.76	+1.04	+0.26	−0.29

The first records how far the streams drift apart. Disagreement between streams is near zero at the first loop depth and rises to about 0.40 by the fourth, so the streams do separate rather than collapse. The extension is therefore active. The second measurement relates that disagreement to bias item by item. Across loop depths the correlation between stream disagreement and the per-item effect size stays below 0.17 in absolute value, and none of the correlations reaches significance at the 0.05 level. The streams differentiate, but where they differentiate is unrelated to which items carry a stereotype preference.

For comparison, representational similarity inside LoopedBERT falls steadily with depth distance: adjacent applications of the shared block remain close, while the first and last differ substantially. The shared block therefore performs different work at different depths, which is the property looped models are designed to have. Taken with the null contrast in Table 3, the reading is that the stereotype reduction observed here follows from sharing weights across depth, and not from how information is routed once shared.

5.6 Training stability

The stream and hyper-connection extension is harder to optimise. At a peak learning rate of 3×10^{-4} , the value that trains the other three encoders without incident, HyperloopBERT reached a validation loss of 2.53 and then rose to 6.86 within one interval before becoming non-finite. Halving the peak learning rate delayed the failure rather than removing it: the model trained past the earlier failure point, reached 2.12, and then diverged again. Both events began shortly after the model’s fastest descent.

ALBERTLoopedBERT, the most heavily shared encoder, also showed one transient loss spike that recovered within a thousand steps, marked in Figure 1. Together the two observations suggest that reusing weights across depth makes optimisation more delicate, and that adding parallel streams on top makes it more delicate still. The snapshots analysed above predate both HyperloopBERT divergences and were checked to hold no non-finite values.

6 Discussion

The direction of the result is consistent across both shared encoders and survives correction, but its size is modest and its scope is narrow. Read carefully, the finding is that at matched language modelling quality an encoder with half the depth-unique weights records a weaker preference for the stereotypical member of a minimal pair. One explanation is that item-level associations are among the first things a model gives up when the same weights must serve several depths, since such associations are narrow and are not needed to

predict text in general. The mechanism is not established here. The evidence is a difference in outcome, not a demonstration of cause.

The negative result carries as much weight as the positive one. HyperloopBERT was included to test whether routing capacity added on top of sharing would change the picture. It did not. For the purpose of reducing stereotype association the extra streams and mixing matrices bought nothing, while making the model harder to train and adding 19 million parameters over LoopedBERT. On this evidence a practitioner choosing between the two on fairness grounds has no reason to prefer the more complex design.

The gate result carries a methodological point. These models give a measurable and separable association signal under pseudo-log-likelihood scoring while answering at chance on coreference. Association-type and behaviour-type instruments therefore do not become usable at the same training scale, which matches the distinction Wang et al. [18] draw between correlation-type benchmarks and those recording differential treatment. A study reporting the first without checking the second could easily overstate what had been shown.

7 Limitations

Each arm is a single seed, so the reported intervals capture variation across benchmark items and not across training runs. The contrasts are paired at item level, which controls item difficulty, but replication over several seeds is needed before the effect size is treated as stable.

The measurement is of the correlation type in the sense of Wang et al. [18]: it records association in a masked language modelling head. It does not show that a downstream classifier built on these encoders would treat groups differently, and because the capability gate of Section 5.3 is not met, the link to downstream behaviour is untested rather than weak. The benchmark also carries documented item-quality problems [2], which is a further reason to read the absolute values as indicative.

The encoders reach roughly two nats of validation loss at the comparison point, well short of a fully trained model. Whether the gap between shared and unshared encoders widens, closes or reverses with longer training is open, and Figure 2 shows effect sizes still rising when the comparison was made.

The evaluation is English-only. An India-centric instrument covering caste and religion [8] was planned, but the mirror available at the time of these experiments could not be used with an English-only tokeniser. Given the regional focus of this venue that omission is stated rather than passed over, and it is the first extension this work would pursue.

8 Conclusion

This work compared four transformer encoders at matched validation loss to ask whether sharing weights across depth changes what a masked language model records about social groups. Both weight-shared encoders showed a lower stereotype effect size than the unshared baseline over 1508 paired items, and both contrasts survived Holm correction. Adding parallel streams and hyper-connections on top of looping gave no further reduction and made training less stable, which is reported here as a negative result. Three cautions attach to the finding. The effects are small, the study uses one seed,

and the models answer at chance on a coreference instrument. It therefore stands as a statement about association in the masked language modelling head, not about downstream behaviour.

Two directions follow. The first is replication across seeds and to lower validation loss, since the association was still growing when the comparison was made. The second is extension to caste and religion, which matters for deployment in the Indian context and which this evaluation does not cover.

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